



## Lab 8 Build a deep neural network (Multilayer Perceptron)

```
In [3]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Input

data = load_digits()
X, y = data.data, data.target

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = Sequential()
model.add(Input(shape=(64,)))
model.add(Dense(128, activation='relu'))
model.add(Dense(64, activation='relu'))
model.add(Dense(10, activation='softmax'))

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

history = model.fit(X_train, y_train, epochs=20, batch_size=32)

loss, accuracy = model.evaluate(X_test, y_test)
print("Accuracy:", accuracy)

plt.plot(history.history['accuracy'])
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.title('Training Accuracy')
plt.show()
```

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

Epoch 1/20

**45/45**  **1s** 3ms/step - accuracy: 0.6298 - loss: 1.4196

Epoch 2/20

**45/45**  **0s** 3ms/step - accuracy: 0.9283 - loss: 0.3698

Epoch 3/20

**45/45**  **0s** 3ms/step - accuracy: 0.9624 - loss: 0.1720

Epoch 4/20

**45/45**  **0s** 3ms/step - accuracy: 0.9798 - loss: 0.1036

Epoch 5/20

**45/45**  **0s** 3ms/step - accuracy: 0.9889 - loss: 0.0699

Epoch 6/20

**45/45**  **0s** 3ms/step - accuracy: 0.9958 - loss: 0.0486

Epoch 7/20

**45/45**  **0s** 3ms/step - accuracy: 0.9972 - loss: 0.0348

Epoch 8/20

**45/45**  **0s** 3ms/step - accuracy: 0.9979 - loss: 0.0270

Epoch 9/20

**45/45**  **0s** 3ms/step - accuracy: 1.0000 - loss: 0.0197

Epoch 10/20

**45/45**  **0s** 3ms/step - accuracy: 0.9993 - loss: 0.0155

Epoch 11/20

**45/45**  **0s** 3ms/step - accuracy: 1.0000 - loss: 0.0129

Epoch 12/20

**45/45**  **0s** 3ms/step - accuracy: 1.0000 - loss: 0.0102

Epoch 13/20

**45/45**  **0s** 4ms/step - accuracy: 1.0000 - loss: 0.0084

Epoch 14/20

**45/45**  **0s** 4ms/step - accuracy: 1.0000 - loss: 0.0074

Epoch 15/20

**45/45**  **0s** 3ms/step - accuracy: 1.0000 - loss: 0.0061

Epoch 16/20

**45/45**  **0s** 3ms/step - accuracy: 1.0000 - loss: 0.0052

Epoch 17/20

**45/45**  **0s** 3ms/step - accuracy: 1.0000 - loss: 0.0046

Epoch 18/20

**45/45**  **0s** 3ms/step - accuracy: 1.0000 - loss: 0.0040

Epoch 19/20

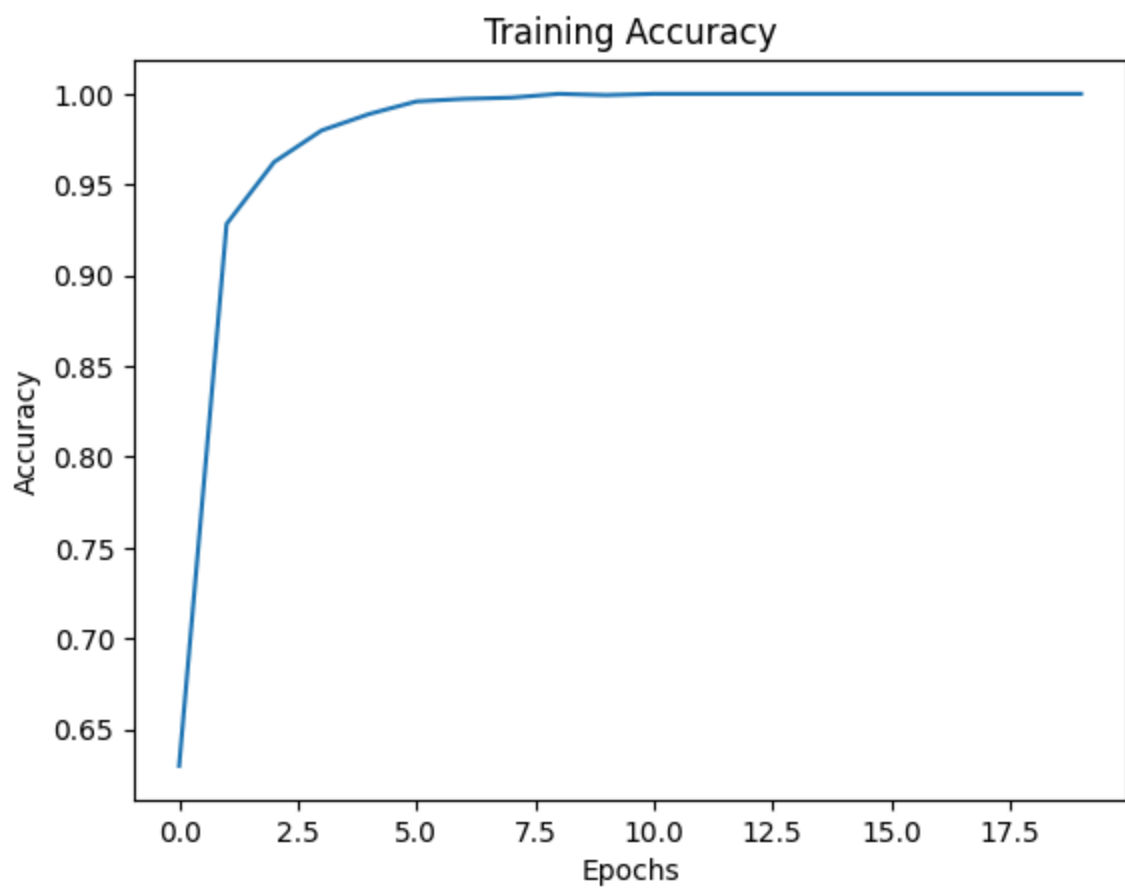
**45/45**  **0s** 4ms/step - accuracy: 1.0000 - loss: 0.0036

Epoch 20/20

**45/45**  **0s** 3ms/step - accuracy: 1.0000 - loss: 0.0032

**12/12**  **0s** 6ms/step - accuracy: 0.9778 - loss: 0.0845

Accuracy: 0.9777777791023254



In [ ]:

In [ ]: